

The Scientific Method

This is a useful sequence, not a fixed recipe. Real investigations loop: results send you back to the question, the design or the hypothesis, and much of science does not involve an experiment at all.

Name _____ Date _____

- ① **Ask a question** Specific, measurable and testable.

- ② **Background research** What is already known? Note your sources.

- ③ **Form a hypothesis** A testable statement that could be shown wrong.

- ④ **Design the investigation** Name the independent, dependent and controlled variables.

- ⑤ **Collect data** Repeat trials. Record units and uncertainty.

- ⑥ **Analyze the data** Tables, graphs, averages. Look for patterns.

- ⑦ **Draw a conclusion** Do the data support the hypothesis, or not? Either result is a real result.

- ⑧ **Communicate** Report what you did and what you found; ask the next question.
